

Microeconomics

Lecture 10: General Review

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Preferences and MRS

Marginal utility

$$MU_x = \frac{\partial U}{\partial x}, \quad MU_y = \frac{\partial U}{\partial y}$$

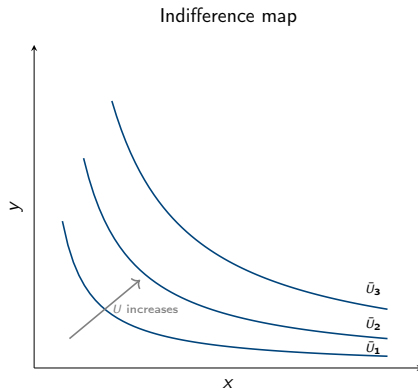
Marginal rate of substitution

$$MRS_{xy} = -\frac{dy}{dx} \Big|_{\bar{U}} = \frac{MU_x}{MU_y}$$

Diminishing MRS along the curve $\Leftrightarrow$ convex preferences.

Three types of preferences

Cobb-Douglas	$U = x^\alpha y^\beta$
Perfect substitutes	$U = ax + by$
Perfect complements	$U = \min\{ax, by\}$



Budget constraint and optimal choice

Budget constraint

$$p_x x + p_y y = M$$

Slope = $-p_x/p_y$ (market trade-off).

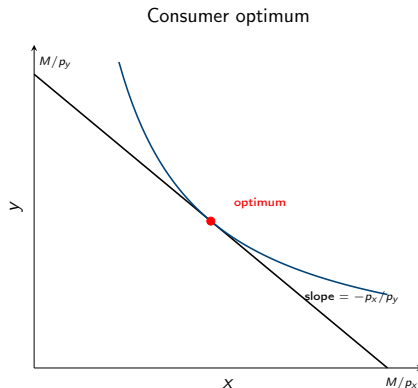
Optimality condition (interior)

$$MRS = \frac{p_x}{p_y}$$

Equivalently: $\frac{MU_x}{p_x} = \frac{MU_y}{p_y}$

Cobb-Douglas solution $U = x^\alpha y^\beta$

$$x^* = \frac{\alpha}{\alpha + \beta} \cdot \frac{M}{p_x}, \quad y^* = \frac{\beta}{\alpha + \beta} \cdot \frac{M}{p_y}$$



The consumer's Lagrangian

Problem

$$\max_{x,y} U(x,y) \quad \text{s.t.} \quad p_x x + p_y y = M$$

Lagrangian and first-order conditions

$$\mathcal{L} = U(x,y) - \lambda(p_x x + p_y y - M)$$

$$\frac{\partial \mathcal{L}}{\partial x} = MU_x - \lambda p_x = 0 \Rightarrow MU_x = \lambda p_x$$

$$\frac{\partial \mathcal{L}}{\partial y} = MU_y - \lambda p_y = 0 \Rightarrow MU_y = \lambda p_y$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(p_x x + p_y y - M) = 0$$

Interpretation of λ

$\lambda = MU_x/p_x = MU_y/p_y$ is the **marginal utility of income**: how much U increases for each additional euro of budget.

Demand and comparative statics

Individual demand curve

Relationship $x^D = x^D(p_x; p_y, M)$ obtained by varying p_x and solving the optimization problem.

Income-consumption curve (ICC) & Engel

Varying M with p_x, p_y fixed $\rightarrow$ ICC.
Projection onto $(M, x) \rightarrow$ Engel curve.

Slutsky equation

$$\underbrace{\frac{\partial x^D}{\partial p_x}}_{\text{total effect}} = \underbrace{\frac{\partial x^c}{\partial p_x}}_{\leq 0 \text{ (substitution)}} - \underbrace{x^D \frac{\partial x^D}{\partial M}}_{\text{income effect}}$$

Classification of goods

Type	Characteristic
Normal	$\partial x^D / \partial M > 0$
Inferior	$\partial x^D / \partial M < 0$
Giffen	income eff. > substitution
Substitutes	$\partial x^D / \partial p_y > 0$
Complements	$\partial x^D / \partial p_y < 0$

Elasticity of demand

Three fundamental elasticities

$$\varepsilon_P = \frac{\partial Q}{\partial P} \cdot \frac{P}{Q} \quad \varepsilon_M = \frac{\partial Q}{\partial M} \cdot \frac{M}{Q} \quad \varepsilon_{xy} = \frac{\partial Q_x}{\partial P_y} \cdot \frac{P_y}{Q_x}$$

Classification of ε_P

$ \varepsilon_P > 1$	elastic
$ \varepsilon_P < 1$	inelastic
$ \varepsilon_P = 1$	unit elastic

Classification of ε_M

$\varepsilon_M < 0$	inferior good
$\varepsilon_M > 0$	normal
$0 < \varepsilon_M < 1$	necessity
$\varepsilon_M > 1$	luxury

Elasticity and total revenue

$$TR = P \cdot Q$$

If $|\varepsilon_P| > 1$: $P \uparrow \Rightarrow TR \downarrow$

If $|\varepsilon_P| < 1$: $P \uparrow \Rightarrow TR \uparrow$

TR is maximized when $|\varepsilon_P| = 1$

Classification of ε_{xy}

$\varepsilon_{xy} > 0$	substitutes
$\varepsilon_{xy} < 0$	complements
$\varepsilon_{xy} = 0$	independent

Production function and marginal products

Production function

$$Q = f(L, K)$$

Marginal and average product

$$MP_L = \frac{\partial Q}{\partial L}, \quad MP_K = \frac{\partial Q}{\partial K}$$

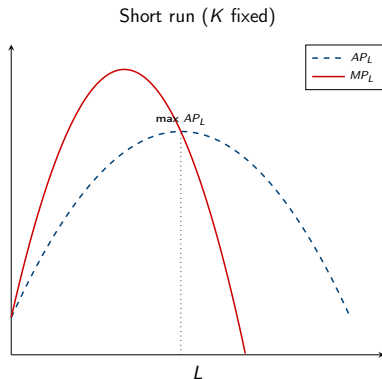
$$AP_L = \frac{Q}{L}, \quad AP_K = \frac{Q}{K}$$

Cobb-Douglas $Q = AL^\alpha K^\beta$

$$MP_L = \alpha AL^{\alpha-1} K^\beta, \quad MP_K = \beta AL^\alpha K^{\beta-1}$$

MP-AP rule

$MP_L > AP_L \Rightarrow AP_L$ rises. $MP_L < AP_L \Rightarrow AP_L$ falls. $MP_L = AP_L \Rightarrow AP_L$ at its maximum.



Returns to scale

Formal definition

Multiply all inputs by $t > 1$:

$$f(tL, tK) \begin{cases} > t f(L, K) & \text{increasing returns (IRS)} \\ = t f(L, K) & \text{constant returns (CRS)} \\ < t f(L, K) & \text{decreasing returns (DRS)} \end{cases}$$

Cobb-Douglas: quick test

$Q = AL^\alpha K^\beta$: substituting tL, tK gives $t^{\alpha+\beta} Q$.

$$\alpha + \beta > 1 \Rightarrow \text{IRS} \quad \alpha + \beta = 1 \Rightarrow \text{CRS} \quad \alpha + \beta < 1 \Rightarrow \text{DRS}$$

Link to costs

IRS $\Leftrightarrow$ economies of scale (AC decreasing in the long run).

DRS $\Leftrightarrow$ diseconomies of scale (AC increasing in the long run).

Isoquants and MRTS

Isoquant

Locus of points (L, K) that produce the same output $\bar{Q}$:

$$\{(L, K) : f(L, K) = \bar{Q}\}$$

MRTS

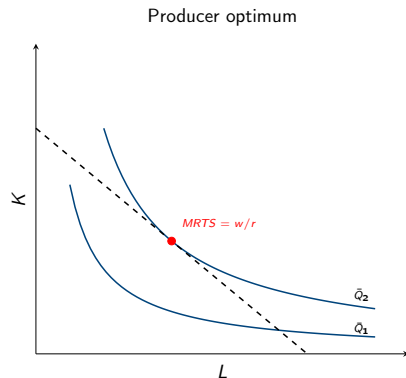
$$MRTS_{LK} = -\frac{dK}{dL} \Big|_{\bar{Q}} = \frac{MP_L}{MP_K}$$

How much K you can give up when L increases by 1, holding Q constant.

Isocost

$$wL + rK = C \quad \Rightarrow \quad K = \frac{C}{r} - \frac{w}{r}L$$

Slope = $-w/r$



Cost minimization: condition and Lagrangian

Problem

$$\min_{L,K} wL + rK \quad \text{s.t.} \quad f(L, K) = \bar{Q}$$

Optimality condition

$$\boxed{MRTS_{LK} = \frac{w}{r}} \Leftrightarrow \frac{MP_L}{w} = \frac{MP_K}{r}$$

The last euro spent on each input produces the same additional quantity of output.

Lagrangian

$$\mathcal{L} = wL + rK - \lambda(f(L, K) - \bar{Q})$$

FOC:

$$w = \lambda MP_L, \quad r = \lambda MP_K$$

$\lambda = w/MP_L = r/MP_K$: marginal cost of producing one additional unit.

Cobb-Douglas: optimal input demands

With $Q = L^\alpha K^\beta$, w, r given:

$$L^* = \left(\frac{\alpha}{\beta} \cdot \frac{r}{w} \right)^{\beta/(\alpha+\beta)} Q^{1/(\alpha+\beta)}, \quad K^* = \left(\frac{\beta}{\alpha} \cdot \frac{w}{r} \right)^{\alpha/(\alpha+\beta)} Q^{1/(\alpha+\beta)}$$

Short-run cost structure

Fundamental definitions

$$TC(Q) = FC + VC(Q)$$

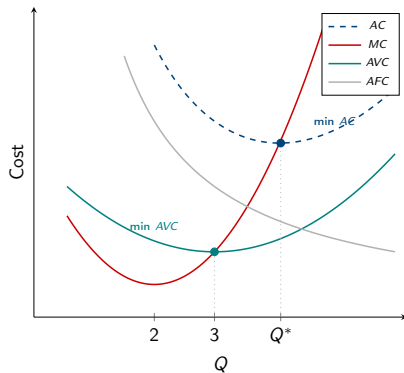
$$AC(Q) = \frac{TC}{Q} = AFC + AVC$$

$$AFC(Q) = \frac{FC}{Q} \quad (\text{always decreasing})$$

$$AVC(Q) = \frac{VC(Q)}{Q}$$

$$MC(Q) = \frac{dTC}{dQ} = \frac{dVC}{dQ}$$

SR cost curves ($TC = \frac{1}{3}Q^3 - 2Q^2 + 5Q + 12$)



MC-AC relationship

$MC < AC \Rightarrow AC$ falls. $MC > AC \Rightarrow AC$ rises. $MC = AC \Rightarrow AC$ at its minimum.

The same rule applies to MC and AVC .

Minimum efficient scale and cost shocks

Minimum efficient scale (MES)

The output level Q^* that minimizes $AC(Q)$:

$$\frac{dAC}{dQ} = 0 \Leftrightarrow MC(Q^*) = AC(Q^*)$$

Example: $TC = F + cQ^2 \Rightarrow Q^* = \sqrt{F/c}$

Effect of shocks

$F \uparrow$ $AC \uparrow$, MC unchanged, $Q^* \uparrow$
 $c \uparrow$ $AC \uparrow$, $MC \uparrow$, $Q^* \downarrow$

Rule of thumb

An increase in **fixed costs** shifts AC up without affecting MC : the firm's supply curve does not change in the short run.

An increase in **variable costs** shifts both AC and MC up.

Sunk costs

They do not affect future decisions: they have already been incurred and cannot be recovered, regardless of the current choice.

Long-run costs

All inputs are variable

In the long run the firm also chooses K , solving the cost-minimization problem for each level of Q .

The LAC curve: envelope of the SR curves

$$LAC(Q) = \min_K AC_{SR}(Q; K)$$

LAC is the **lower envelope** of all short-run AC curves (one for each fixed K).

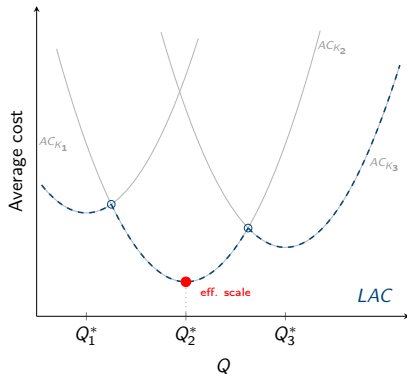
Economies and diseconomies of scale

LAC decreasing $\Leftrightarrow$ **economies of scale** (IRS).

LAC increasing $\Leftrightarrow$ **diseconomies of scale** (DRS).

Minimum of LAC = long-run efficient scale.

LAC = lower envelope of SR AC curves



Key formulas

Concept	Formula	Optimality condition
Consumer optimum	$MRS = MU_x/MU_y$	$MRS = p_x/p_y$
Cobb-Douglas demand	$x^* = \frac{\alpha}{\alpha+\beta} \cdot \frac{M}{p_x}$	constraint $p_x x + p_y y = M$
Slutsky	$\frac{\partial x^D}{\partial p_x} = \frac{\partial x^c}{\partial p_x} - x^D \frac{\partial x^D}{\partial M}$	subst. effect ≤ 0
Price elasticity	$\varepsilon_P = \frac{\partial Q}{\partial P} \cdot \frac{P}{Q}$	TR max if $ \varepsilon_P = 1$
Marginal product	$MP_L = \partial Q / \partial L$	MP-AP rule
MRTS	MP_L / MP_K	$MRTS = w/r$
Returns to scale	$f(tL, tK)$ vs $t f(L, K)$	$\alpha + \beta$ for C-D
Marginal cost	$MC = dTC/dQ$	$MC = AC$ at min AC
Minimum efficient scale	$TC = F + cQ^2$	$Q^* = \sqrt{F/c}$
LR cost min.	$MRTS = w/r$	LAC = envelope

Graphs you must be able to draw

Consumer theory

- Indifference curve (Cobb-D., substitutes, complements)
- Budget line and optimum (tangency)
- Price-consumption curve (PCC) and demand
- Income-consumption curve (ICC) and Engel
- Slutsky decomposition

Elasticity

- Demand curve and elasticity classification
- TR vs $|\varepsilon_P|$ (bell shape)

Technology

- TP , AP_L , MP_L curves (short run)
- Isoquants (C-D, substitutes, complements)
- Isoquant + isocost $\rightarrow$ producer optimum

Costs

- TC , FC , VC curves
- AC , AVC , AFC , MC curves (U-shaped)
- Shifts from shocks to F and c
- LAC as envelope of the SR curves

Exercise 1

A consumer has preferences represented by

$$U(x, y) = \sqrt{x} + y$$

with prices $p_x = 2$, $p_y = 1$ and income $M = 10$.

- (a) Write the Lagrangian and derive the first-order conditions.
- (b) Find the optimal bundle (x^*, y^*) . *Watch out for a corner solution.*
- (c) Compute the *MRS* at the optimum and check the tangency condition.
- (d) Income rises to $M' = 15$. How does the demand for x change? And that for y ? What kind of goods are x and y ?
- (e) Now suppose $M'' = 0.05$. Find the optimum. Does the standard procedure (interior FOCs) still work? Why?

Solution (a)

(a) Lagrangian

$$\mathcal{L}(x, y, \lambda) = \sqrt{x} + y - \lambda(2x + y - 10)$$

First-order conditions

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{1}{2\sqrt{x}} - 2\lambda = 0 \Rightarrow \lambda = \frac{1}{4\sqrt{x}}$$

$$\frac{\partial \mathcal{L}}{\partial y} = 1 - \lambda = 0 \Rightarrow \lambda = 1$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(2x + y - 10) = 0 \Rightarrow 2x + y = 10$$

Note

$\lambda = 1$ from the FOC for y : the marginal utility of income is 1, constant in y . This is the distinctive feature of quasi-linear preferences.

Solution (b)

(b) Obtain x^* by combining the two FOCs

$$\frac{1}{4\sqrt{x}} = 1 \Rightarrow \sqrt{x} = \frac{1}{4} \Rightarrow \boxed{x^* = \frac{1}{16}}$$

Obtain y^* from the constraint

$$2 \cdot \frac{1}{16} + y^* = 10 \Rightarrow y^* = 10 - \frac{1}{8} = \boxed{\frac{79}{8} \approx 9.875}$$

Check: interior solution

$x^* = 1/16 > 0$ and $y^* = 79/8 > 0$: the solution is interior, not a corner. ✓

The constraint binds: $2 \cdot \frac{1}{16} + \frac{79}{8} = \frac{1}{8} + \frac{79}{8} = \frac{80}{8} = 10$. ✓

When would there be a corner solution?

If $M < p_x \cdot x^* = 2 \cdot \frac{1}{16} = \frac{1}{8}$, the consumer would spend everything on x and $y^* = 0$. With $M = 10$ we are far from that case.

Solutions (c) and (d)

(c) Checking the tangency condition

$$MRS = \frac{MU_x}{MU_y} = \frac{\frac{1}{2\sqrt{x}}}{1} = \frac{1}{2\sqrt{1/16}} = \frac{1}{2 \cdot \frac{1}{4}} = 2 = \frac{p_x}{p_y} \checkmark$$

(d) Effect of $M' = 15$

The demand for x *does not depend on* M : from the FOC, $x^* = \frac{1}{16}$ regardless of income.

The demand for y absorbs all the additional income:

$$y^{*'} = \frac{M' - 2x^*}{1} = 15 - \frac{1}{8} = \frac{119}{8} \approx 14.875$$

Classification

x is an **income-neutral** good (a peculiar case of quasi-linear preferences: $\partial x^* / \partial M = 0$, $\varepsilon_M^x = 0$).

y is a **normal** good with $\partial y^* / \partial M = 1$ (it fully absorbs every additional euro);

$\varepsilon_M^y = M / y^* = 80 / 79 \approx 1.01$.

Solution (e)

(e) Case $M'' = 0.05$

Let us naively apply the interior FOCs: $x^* = 1/16 = 0.0625$. But the constraint would require

$$y^* = M'' - 2x^* = 0.05 - 2 \times 0.0625 = 0.05 - 0.125 = -0.075 < 0$$

Problem: $y^* < 0$ is not feasible

The interior solution requires income to be sufficient to finance

$$p_x x^* = 2 \times \frac{1}{16} = \frac{1}{8} = 0.125 > M'' = 0.05.$$

With $M'' < p_x x^{\text{int}} = \frac{1}{8}$, the consumer **cannot** reach the optimal interior bundle. The corner solution with $y^* = 0$ is binding.

Corner optimum: $y^* = 0$, everything spent on x

$$p_x x^* = M'' \Rightarrow x^* = \frac{M''}{p_x} = \frac{0.05}{2} = \boxed{0.025}$$

Solution (e)

Check: the MRS exceeds the relative price at the corner

$$MRS(0.025, 0) = \frac{1}{2\sqrt{0.025}} = \frac{1}{2 \times 0.158} \approx 3.16 > 2 = \frac{p_x}{p_y}$$

The consumer *would like* even more x , but the budget is exhausted: x is subjectively worth more than it costs, but there is no income left. Correct corner condition: $MRS \geq p_x/p_y$ with $y^* = 0$.

Exercise 2

A consumer's **individual** demand for good x is

$$x^D(p_x, M) = \frac{M}{4 p_x}$$

The consumer has income $M_0 = 200$ and faces the price $p_0 = 5$.

- (a) Compute x_0 , the price elasticity ε_P and the income elasticity ε_M at the point (p_0, M_0) .
- (b) Classify the good and determine what happens to the seller's total revenue if p_x rises by 10%.
- (c) Using the Slutsky equation, decompose $\partial x^D / \partial p_x$ into substitution effect and income effect. What is the sign of each effect?
- (d) Suppose p_x falls from 5 to 4. Compute the *exact* change in x^D and compare it with the elasticity-based approximation.

Solution (a)

(a) Initial quantity

$$x_0 = \frac{200}{4 \times 5} = \frac{200}{20} = \boxed{10}$$

Price elasticity

$$\frac{\partial x^D}{\partial p_x} = -\frac{M}{4 p_x^2} \Rightarrow \varepsilon_P = -\frac{M}{4 p_x^2} \cdot \frac{p_x}{x^D} = -\frac{M}{4 p_x} \cdot \frac{4 p_x}{M} = \boxed{-1}$$

Income elasticity

$$\frac{\partial x^D}{\partial M} = \frac{1}{4 p_x} \Rightarrow \varepsilon_M = \frac{1}{4 p_x} \cdot \frac{M}{x^D} = \frac{1}{4 p_x} \cdot \frac{M \cdot 4 p_x}{M} = \boxed{1}$$

Note

$\varepsilon_P = -1$ and $\varepsilon_M = 1$ for every value of p_x and M : they are constant. This is the distinctive feature of unit-elastic demand, typical of Cobb-Douglas: indeed $x^D = M/(4p_x)$ is the Cobb-Douglas demand with expenditure share $1/4$.

Solutions (b) and (c)

(b) Classification and total revenue

- $|\varepsilon_P| = 1$: **unit** elasticity. Total revenue $TR = p_x \cdot x^D = M/4$ is *constant* as p_x varies: a 10% price increase leaves TR unchanged.
- $\varepsilon_M = 1 > 0$: **normal** good (not a luxury: $\varepsilon_M = 1$, not > 1).

(c) Slutsky decomposition

$$\underbrace{\frac{\partial x^D}{\partial p_x}}_{=-M/(4p_x^2)} = \underbrace{\frac{\partial x^c}{\partial p_x}}_{\leq 0} - \underbrace{x^D \cdot \frac{\partial x^D}{\partial M}}_{\geq 0}$$

Numerical computation at $(p_0, M_0) = (5, 200)$, $x_0 = 10$:

$$\frac{\partial x^D}{\partial p_x} = -\frac{200}{4 \cdot 25} = -2, \quad x^D \cdot \frac{\partial x^D}{\partial M} = 10 \cdot \frac{1}{20} = \frac{1}{2}$$

$$\frac{\partial x^c}{\partial p_x} = -2 + \frac{1}{2} = \boxed{-\frac{3}{2}} < 0 \checkmark, \quad \text{income effect} = -\frac{1}{2} < 0$$

Both negative: x is a normal good (income and substitution effects work in the same direction).

Solution (d)

(d) Exact change: p_x falls from 5 to 4

$$x^D(4, 200) = \frac{200}{4 \times 4} = \frac{200}{16} = 12.5$$

$$\Delta x^{\text{exact}} = 12.5 - 10 = \boxed{+2.5}$$

Elasticity-based approximation

Percentage change in price: $\hat{p} = (4 - 5)/5 = -20\%$.

$$\hat{x} \approx \varepsilon_P \cdot \hat{p} = (-1) \times (-20\%) = +20\%$$

$$\Delta x^{\text{approx}} \approx 10 \times 20\% = \boxed{+2.0}$$

Comparison

The approximation (+2) underestimates the exact change (+2.5) because the derivative is evaluated at the initial point: as p_x falls, $|\partial x^D / \partial p_x|$ grows (the demand curve is not linear). The discrepancy increases with the size of the shock.

Exercise 3

A firm has the production function

$$Q = f(L, K) = L^{1/3}K^{2/3}$$

In the short run $K = \bar{K} = 8$ is fixed.

- (a) Write the short-run production function $Q(L; \bar{K} = 8)$ and compute MP_L and AP_L .
- (b) Show that $MP_L < AP_L$ for every $L > 0$, and interpret the result.
- (c) The firm increases L from 8 to 27. Compute the discrete change in Q and compare it with the approximation based on MP_L .
- (d) Check the returns to scale of the function $Q = L^{1/3}K^{2/3}$ in the long run.
- (e) In the short run (with $K = 8$ fixed) you showed that MP_L is decreasing. Does this contradict the constant returns to scale of part (d)? Explain the conceptual distinction.

Solution (a)

(a) SR function with $K = 8$

$$Q(L) = L^{1/3} \cdot 8^{2/3} = L^{1/3} \cdot 4 = 4L^{1/3}$$

Marginal product of labor

$$MP_L = \frac{dQ}{dL} = 4 \cdot \frac{1}{3} L^{-2/3} = \boxed{\frac{4}{3L^{2/3}}}$$

Average product of labor

$$AP_L = \frac{Q}{L} = \frac{4L^{1/3}}{L} = \boxed{\frac{4}{L^{2/3}}}$$

Solution (b)

(b) Comparing MP_L and AP_L

$$\frac{MP_L}{AP_L} = \frac{4/(3L^{2/3})}{4/L^{2/3}} = \frac{1}{3} < 1 \quad \forall L > 0$$

Hence $MP_L = \frac{1}{3} AP_L < AP_L$ for every $L > 0$.

Interpretation

The marginal product is always below the average: each additional worker contributes less than the average of the existing workforce, so AP_L is *always* decreasing. This reflects diminishing marginal returns to labor in the short run (with K fixed).

Solution (c)

(c) Discrete change: L from 8 to 27

$$Q(8) = 4 \cdot 8^{1/3} = 4 \cdot 2 = 8, \quad Q(27) = 4 \cdot 27^{1/3} = 4 \cdot 3 = 12$$

$$\Delta Q = 12 - 8 = 4$$

Approximation with MP_L evaluated at $L = 8$:

$$MP_L(8) = \frac{4}{3 \cdot 8^{2/3}} = \frac{4}{3 \cdot 4} = \frac{1}{3}, \quad \Delta Q \approx MP_L(8) \cdot \Delta L = \frac{1}{3} \cdot 19 = 6.3$$

Why is the approximation so far off?

$\Delta L = 19$ is a *very* large discrete change; the derivative is a good approximation only for $\Delta L \rightarrow 0$. Here it overestimates: MP_L decreases along the way.

Solution (d)

(d) Returns-to-scale test

Multiply both inputs by $t > 1$:

$$\begin{aligned} f(tL, tK) &= (tL)^{1/3} (tK)^{2/3} = t^{1/3} t^{2/3} L^{1/3} K^{2/3} \\ &= t^{1/3+2/3} f(L, K) = t^1 \cdot f(L, K) \end{aligned}$$

Conclusion

$\alpha + \beta = \frac{1}{3} + \frac{2}{3} = 1$: the technology exhibits **constant returns to scale** (CRS).
Doubling all inputs exactly doubles output.

Link to costs

With CRS, the LAC curve is flat: there are neither economies nor diseconomies of scale in the long run. The optimal size of the firm is indeterminate (technically any scale is efficient).

Solution (e)

(e) Diminishing marginal returns $\neq$ decreasing returns to scale

Short run: one input is fixed

Decreasing MP_L means: increasing *only* L (with K fixed), each additional unit of labor contributes *less* to output.

It is a **one-dimensional** property: we move along a path on which K does not change.

Long run: all inputs vary

Returns to scale describe what happens when all inputs are increased *proportionally*:
 $f(tL, tK)$ vs $t f(L, K)$.

It is a **radial** property: we move along a ray from the starting point.

They can coexist without contradiction

With $Q = L^{1/3} K^{2/3}$:

- $\partial^2 Q / \partial L^2 = -\frac{2}{9} L^{-5/3} K^{2/3} < 0$: MP_L **decreasing** in L (SR, K fixed) ✓
- $f(tL, tK) = t \cdot f(L, K)$: **constant** returns to scale (LR) ✓

A function can have diminishing marginal returns in each input *and* constant or increasing returns to scale: they are orthogonal concepts.

Exercise 4

A firm produces with a perfect-complements technology:

$$Q = \min\{L, 2K\}$$

The wage is $w = 4$ and the cost of capital is $r = 3$.

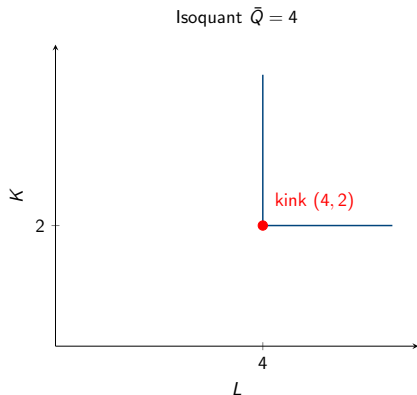
- (a) Draw (qualitatively) the isoquant for $\bar{Q} = 4$. Where is the kink?
- (b) Find the optimal input demands $L^*(Q)$ and $K^*(Q)$ that minimize cost.
Why does the $MRTS = w/r$ method not apply directly here?
- (c) Write the total cost function $C(Q)$ and compute MC and AC .
- (d) Input prices change: $w' = 2$, $r' = 6$. How do $C(Q)$, MC and the optimal input choice change?

Solution (a)

(a) Isoquant for $\bar{Q} = 4$

The technology requires $L = \bar{Q}$ and $2K = \bar{Q}$, i.e., $L \geq 4$ and $K \geq 2$. The isoquant is an **L-shaped curve** with its kink at:

$$\bar{Q} = \min\{L, 2K\} = 4 \Rightarrow L = 4, K = 2 \quad (\text{kink})$$



Why $MRTS = w/r$ does not work

The $MRTS$ is not defined at the kink (the function is not differentiable there). Along the horizontal and vertical segments the $MRTS$ is 0 or $+\infty$. The tangency condition cannot be applied.

Solutions (b) and (c)

(b) Optimal inputs

Any unit of L in excess of $\bar{Q}$ or of K in excess of $\bar{Q}/2$ is wasted. Cost is minimized exactly at the kink:

$$L^*(Q) = Q, \quad K^*(Q) = \frac{Q}{2}$$

(c) Total cost function

$$C(Q) = wL^* + rK^* = 4Q + 3 \cdot \frac{Q}{2} = 4Q + \frac{3}{2}Q = \frac{11}{2}Q$$

$$MC = \frac{dC}{dQ} = 5.5 \quad (\text{constant}), \quad AC = \frac{C(Q)}{Q} = 5.5 \quad (\text{constant})$$

Note

With CRS and perfect complements, $MC = AC$ is constant for every Q . There is no preferred efficient scale: any Q is equally efficient.

Solution (d)

(d) New prices: $w' = 2$, $r' = 6$

The optimal input demands *do not change*: $L^* = Q$ and $K^* = Q/2$ depend only on the technology, not on prices (with perfect complements substitution is impossible).

New cost function

$$C'(Q) = 2Q + 6 \cdot \frac{Q}{2} = 2Q + 3Q = \boxed{5Q}$$

$$MC' = AC' = \boxed{5} < 5.5$$

Interpretation

The fall in w (from 4 to 2) reduces the cost of the more intensively used input ($L^* = Q > K^* = Q/2$), lowering total cost despite the increase in r . The firm keeps using exactly the same combination (L^*, K^*): with perfect complements the technical mix is rigid.

Exercise 5

In the short run a firm has total cost:

$$TC(Q) = Q^3 - 6Q^2 + 15Q + 48$$

- (a) Identify FC , $VC(Q)$, $AFC(Q)$, $AVC(Q)$, $AC(Q)$ and $MC(Q)$.
- (b) Find the Q^* that minimizes $AC(Q)$. Check that $MC(Q^*) = AC(Q^*)$.
- (c) Find the Q^{**} that minimizes $AVC(Q)$ and compute $AVC(Q^{**})$.
- (d) In the long run the firm can choose among three plants: small, $TC_1 = Q^3 - 4Q^2 + 14Q + 27$; the current one ($TC_2 = TC$); large, $TC_3 = Q^3 - 10Q^2 + 27Q + 75$. Construct the LAC and determine which plant is efficient for low, medium and high levels of Q .

Solution (a)

(a) Cost components

$$FC = 48, \quad VC(Q) = Q^3 - 6Q^2 + 15Q$$

$$AFC(Q) = \frac{48}{Q}, \quad AVC(Q) = Q^2 - 6Q + 15$$

$$AC(Q) = AFC + AVC = Q^2 - 6Q + 15 + \frac{48}{Q}$$

$$MC(Q) = \frac{dTC}{dQ} = 3Q^2 - 12Q + 15$$

Remark

$AVC(Q) = Q^2 - 6Q + 15 = (Q - 3)^2 + 6 \geq 6$ for every Q : AVC is always positive, with its minimum at $Q = 3$.

Solution (b)

(b) Condition $MC = AC$

$$3Q^2 - 12Q + 15 = Q^2 - 6Q + 15 + \frac{48}{Q}$$

$$2Q^2 - 6Q = \frac{48}{Q}$$

$$2Q^3 - 6Q^2 = 48 \Rightarrow Q^3 - 3Q^2 - 24 = 0$$

Solution (b)

Method: direct derivative of AC

$$\frac{d AC}{dQ} = 2Q - 6 - \frac{48}{Q^2} = 0 \Rightarrow 2Q^3 - 6Q^2 - 48 = 0 \Rightarrow Q^3 - 3Q^2 - 24 = 0$$

There is no integer root. To locate Q^* , we evaluate the sign of $MC - AC$ at integer points:

$$Q = 4 : MC - AC = 15 - 19 = -4 < 0 \Rightarrow MC < AC \Rightarrow AC \text{ still falling}$$

$$Q = 5 : MC - AC = 30 - 19.6 = +10.4 > 0 \Rightarrow MC > AC \Rightarrow AC \text{ rising}$$

$$\Rightarrow \boxed{Q^* \in (4, 5), \quad Q^* \approx 4.3}$$

Solution (c)

(c) Minimum of AVC

$$\frac{d AVC}{dQ} = 2Q - 6 = 0 \Rightarrow \boxed{Q^{**} = 3}$$

$$AVC(3) = 9 - 18 + 15 = \boxed{6}$$

$Q^{**} = 3 < Q^* \approx 4.3$: the minimum of AVC always comes before the minimum of AC .

Why is the minimum of AC always to the right of that of AVC ?

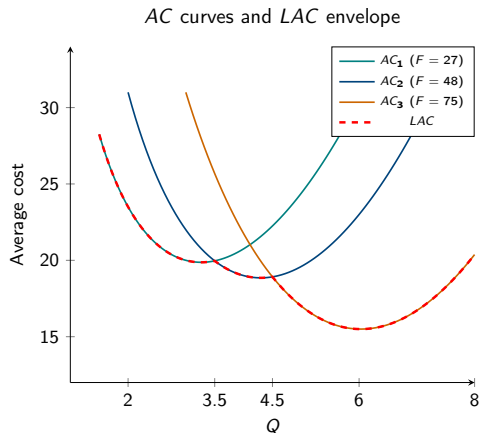
$AC = AVC + AFC$. Since AFC is always decreasing, adding AFC to AVC shifts the minimum of the sum to the right: when AVC is already at its minimum, AFC is still falling and keeps pulling AC down for a further stretch. The minimum point of AC is where the fall in AFC can no longer offset the rise in AVC , i.e., where $MC = AC$.

Solution (d)

(d) Three plants: $F_1 = 27$, $F_2 = 48$, $F_3 = 75$

$AC_1 = (Q - 2)^2 + 10 + \frac{27}{Q}$, $AC_2 = (Q - 3)^2 + 6 + \frac{48}{Q}$, $AC_3 = (Q - 5)^2 + 2 + \frac{75}{Q}$. LAC is the lower envelope:

$$LAC(Q) = \min_{i \in \{1,2,3\}} AC_i(Q)$$



Efficient plant by output range

$$AC_1 = AC_2 \iff 2Q^2 - Q - 21 = 0 \iff Q = 3.5$$

$$AC_2 = AC_3 \iff 4Q^2 - 12Q - 27 = 0 \iff Q = 4.5$$

Hence: **small** plant for $Q < 3.5$; **current** plant for $3.5 \leq Q \leq 4.5$; **large** plant for $Q > 4.5$.

Minimum of LAC on the AC_3 segment:

$$2Q^3 - 10Q^2 - 75 = 0 \Rightarrow Q \approx 6.03, \quad LAC \approx 15.50.$$

Note

Higher F , but lower AVC at high Q . With the same AVC , the plant with the lowest F would dominate at every Q .

Summary of common exam mistakes

Consumer theory

- Forgetting to check for a corner solution when MU_y/p_y is constant (quasi-linear)
- Confusing λ (marginal utility of income) with utility itself

Elasticity

- Forgetting that the substitution effect in the Slutsky equation is always ≤ 0
- Confusing the sign of ε_P with the elastic/inelastic classification (use the absolute value)
- Approximating large changes with the elasticity: the error grows with $|\Delta p|$

Technology and production

- Using the derivative for overly large discrete approximations
- Forgetting that returns to scale cannot be tested with one input fixed

Costs

- Applying $MRTS = w/r$ with perfect complements (not differentiable) or perfect substitutes (corner solutions)
- Confusing the minimum of AVC with the minimum of AC (the latter is always further to the right)
- Forgetting that LAC is the *lower* envelope, not the average of the AC curves